Local Fidelity, Global Alias

A Pipeline Proof of the Three-Dimensional Jacobian Counterexample

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AI-assisted proof construction and exact symbolic audit

> **Central mechanism.** A polynomial representation may preserve every infinitesimal degree of freedom while erasing a discrete global state label. The Jacobian detects the first fact. A collision detects the second.

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Verification: Exact polynomial arithmetic over the rational numbers

Abstract

We give a self-contained proof that the Jacobian conjecture is false in dimension three. The explicit degree-seven map is already publicly known; the contribution of this note is a proof architecture that derives the certificate from a general counterexample-construction pipeline.

The full state is a *marked factorization* (L, Q) of a binary cubic. The visible representation is the product LQ, which forgets which of the cubic's three roots was assigned to the linear factor. A resultant normalization removes the continuous scaling gauge and makes multiplication locally faithful, while the discrete three-way ambiguity survives globally. A specially aligned affine slice then closes under a two-level residual hierarchy. Promoting those residuals to coordinates produces an affine three-space and, after linear coordinate changes, the compact counterexample.

Thus the proof is routed as follows.

marked state $\longrightarrow$ **visible product** $\longrightarrow$ **gauge normalization** $\longrightarrow$ **local fidelity** $\longrightarrow$ **global alias** $\longrightarrow$ **boundary closure** $\longrightarrow$ **finite certificate**

No numerical approximation is used.

1 Theorem intake and proof route

The Jacobian conjecture over $\mathbb{C}$ asserts that every polynomial map

$$F : \mathbb{C}^n \longrightarrow \mathbb{C}^n$$

with nonzero constant Jacobian determinant has a polynomial inverse.

The claim to be proved is existential.

Theorem 1. *There exists a polynomial map $F : \mathbb{C}^3 \to \mathbb{C}^3$ whose Jacobian determinant is the nonzero constant -2, but which is not injective. Consequently, the Jacobian conjecture is false in dimension three and in every dimension $n \geq 3$.*

The proof obligations are exact:

1. construct a polynomial map $F : \mathbb{C}^3 \to \mathbb{C}^3$;
2. prove $\det DF \equiv -2$;
3. exhibit distinct points with the same image.

The pipeline adds four construction obligations before the final certificate.

Pipeline stage	**Mathematical obligation**
Full state and visible representation	Identify what information the map retains and what it erases
Invariant-preserving constraint	Remove trivial continuous nonuniqueness without destroying the target collision
Local control	Prove the constrained representation is infinitesimally faithful
Boundary closure	Find polynomial coordinates covering the degenerate boundary and the generic region simultaneously

2 The marked-factor representation

Let U, V be homogeneous variables. Write

$$L(U,V) = aU + bV, \qquad Q(U,V) = cU^2 + dUV + eV^2.$$

Their product is the binary cubic

$$LQ = ac\,U^3 + (ad+bc)U^2V + (ae+bd)UV^2 + be\,V^3.$$

Therefore multiplication defines the polynomial representation

$$M : \mathbb{C}^5 \longrightarrow \mathbb{C}^4, \qquad M(a,b,c,d,e) = (ac,\ ad+bc,\ ae+bd,\ be).$$

The input remembers a marked decomposition: one root belongs to L, while the other two belong to Q. The output remembers only the product.

If a cubic has three distinct linear factors,

$$C = L_1L_2L_3,$$

then the three marked states

$$(L_1, L_2L_3), \qquad (L_2, L_1L_3), \qquad (L_3, L_1L_2)$$

all have visible representation C. This is the global alias we want, but multiplication also has a trivial continuous ambiguity:

$$(L,Q) \longmapsto (\lambda L, \lambda^{-1}Q), \qquad \lambda \in \mathbb{C}^\times.$$

We remove that gauge before testing whether any genuine collision remains.

3 Gauge normalization and local fidelity

Define the resultant

$$R(L,Q) = R(a,b,c,d,e) = a^2e - abd + b^2c.$$

It vanishes exactly when L and Q share a projective root. Under the gauge transformation above,

$$R(\lambda L, \lambda^{-1} Q) = \lambda R(L, Q).$$

Hence every state with $R \neq 0$ has a unique gauge representative satisfying

$$R = 1.$$

Let

$$Y = \{(a, b, c, d, e) \in \mathbb{C}^5 : R(a, b, c, d, e) = 1\}.$$

Lemma 2 (Multiplication is locally faithful on the normalized state space)**.** *For every $p \in Y$, the differential*

$$d(M|_Y)_p : T_p Y \longrightarrow \mathbb{C}^4$$

is an isomorphism.

Proof. Since $R(p) = 1$, the polynomials L and Q are coprime. Let $(\dot{L}, \dot{Q}) \in T_p Y$ lie in the kernel of dM_p. Differentiating LQ gives

$$\dot{L}\, Q + L\, \dot{Q} = 0.$$

Thus $L \mid \dot{L} Q$. Because $\gcd(L, Q) = 1$, unique factorization in $\mathbb{C}[U, V]$ gives $L \mid \dot{L}$. Both are linear, so $\dot{L} = tL$ for some $t \in \mathbb{C}$. The differentiated product identity then gives $\dot{Q} = -tQ$.

The only possible infinitesimal kernel direction is therefore the gauge direction. But differentiating the resultant scaling law at $\lambda = 1$ gives

$$dR_p(tL, -tQ) = tR(p) = t.$$

Tangency to $Y = \{R = 1\}$ requires $dR_p(\dot{L}, \dot{Q}) = 0$, hence $t = 0$. Thus the differential is injective.

The hypersurface Y is smooth: R is homogeneous of degree three, so Euler's identity gives

$$aR_a + bR_b + cR_c + dR_d + eR_e = 3R = 3$$

on Y; consequently $\nabla R \neq 0$. Hence $\dim T_p Y = 4$, equal to the target dimension. The injective differential is therefore an isomorphism. □

The pipeline control has now passed: after the trivial gauge is removed, multiplication loses no infinitesimal information. Nevertheless, the three distinct marked states constructed above remain. The resultant normalization removes a continuous redundancy but cannot restore the erased discrete label.

Indeed, each marked factorization of a cubic with three distinct roots has nonzero resultant. Choosing $\lambda = R(L, Q)^{-1}$ in the scaling law normalizes it to $R = 1$ without changing the product, so all three marked states survive inside Y.

4 The affine boundary and residual closure

Let $\mathcal{V} \subset \mathbb{C}^4$ be the affine hyperplane where the U^2V-coefficient equals one:

$$\mathcal{V} = \{(f, g, h, i) \in \mathbb{C}^4 : g = 1\}.$$

Its normalized preimage is

$$X = Y \cap M^{-1}(\mathcal{V}) = \left\{ (a, b, c, d, e) \in \mathbb{C}^5 : \begin{array}{l} a^2e - abd + b^2c = 1, \\ ad + bc = 1 \end{array} \right\}.$$

The double-root slice enters here through the middle coefficient $ad + bc$. At the boundary $a = 0$, the two constraints become

$$b^2c = 1, \qquad bc = 1,$$

and therefore

$$b = c = 1.$$

The generic cubic-quadratic intersection does not branch in the affine boundary: it collapses to a single base point in (b, c), with d, e free. This is the closure seam.

The first two normalized residuals are

$$b - 1 = ay, \qquad c - 1 + \frac{3}{2}ay = a^2z.$$

Promoting y and z to coordinates yields the following global chart.

Lemma 3 (Polynomial closure of the slice)**.** *The map $\Phi : \mathbb{C}^3 \to X$, defined by*

$$a = a, \qquad b = 1 + ay, \qquad c = 1 - \frac{3}{2}ay + a^2z,$$

$$d = \frac{1}{2}y - az + \frac{3}{2}ay^2 - a^2yz,$$

$$e = -2z + 4y^2 - 4ayz + 3ay^3 - 2a^2y^2z,$$

is a polynomial isomorphism.

Proof. Substitution in the displayed formulas gives the two polynomial identities

$$ad + bc = 1, \qquad 2bd - ae = y.$$

Because $b = 1 + ay$, these identities imply

$$\begin{aligned} a^2e - abd + b^2c &= a^2e - abd + b(1 - ad) \\ &= a^2e - 2abd + b \\ &= b - ay \\ &= 1. \end{aligned}$$

Thus $\Phi(\mathbb{C}^3) \subseteq X$.

Conversely, take $(a, b, c, d, e) \in X$ and define polynomially

$$y = 2bd - ae, \qquad z = 2d^2 + 3dy + e\left(c - \frac{3}{2}\right).$$

Using the two equations of X,

$$\begin{aligned} ay &= 2abd - a^2e \\ &= abd + b^2c - 1 \\ &= b(ad + bc) - 1 \\ &= b - 1. \end{aligned}$$

Hence $b = 1 + ay$. Next,

$$\begin{aligned} \frac{3}{2}y - d - yc &= d(3b - 1 - 2bc) + ae\left(c - \frac{3}{2}\right) \\ &= a\left(3dy + 2d^2 + e\left(c - \frac{3}{2}\right)\right) \\ &= az, \end{aligned}$$

where we used $bc = 1 - ad$ and $b - 1 = ay$. Since $b = 1 + ay$ and $bc = 1 - ad$,

$$c + ayc = 1 - ad, \qquad \text{so} \qquad c - 1 = -a(d + yc).$$

Together with the preceding identity, this gives

$$c - 1 + \frac{3}{2}ay = a^2z,$$

which recovers the displayed formula for c.

If $a \neq 0$, substituting the recovered b, c into $ad + bc = 1$ gives the displayed formula for d. The relation $y = 2bd - ae$ then gives the displayed formula for e.

It remains to treat the boundary rather than cancel a. If $a = 0$, the boundary equations give $b = c = 1$. The polynomial inverse coordinates then give

$$y = 2d, \qquad z = 8d^2 - \frac{1}{2}e.$$

Therefore

$$d = \frac{1}{2}y, \qquad e = -2z + 4y^2,$$

which are exactly the $a = 0$ cases of the displayed formulas. Thus the two polynomial coordinates in the inverse construction define an inverse to Φ on all of X, including the degenerate fiber. $\square$

This is the differential-closure step of the pipeline: the first residual is divisible by a, the corrected second residual is divisible by a^2, and the resulting coordinates close after two levels.

5 The induced map and the structural Jacobian

Identify $\mathcal{V}$ with $\mathbb{C}^3$ by deleting its fixed second coordinate. The induced map

$$P : \mathbb{C}^3 \longrightarrow \mathbb{C}^3$$

is

$$P(a, y, z) = \big(G_1(a, y, z), G_2(a, y, z), G_3(a, y, z)\big),$$

where

$$G_1 = a - \frac{3}{2}a^2y + a^3z,$$

$$G_2 = \frac{1}{2}y - 3az + 6ay^2 - 6a^2yz + \frac{9}{2}a^2y^3 - 3a^3y^2z,$$

$$G_3 = -2z + 4y^2 - 6ayz + 7ay^3 - 6a^2y^2z + 3a^2y^4 - 2a^3y^3z.$$

These are simply the remaining coefficients

$$(G_1, G_2, G_3) = (ac,\ ae + bd,\ be)$$

after substitution of the polynomial chart.

By Lemma 2, $M|_Y$ is locally invertible. Restricting it to the inverse image of the smooth hyperplane $\mathcal{V}$ preserves local invertibility. Lemma 3 supplies polynomial coordinates on that inverse image. Therefore DP is invertible at every point of $\mathbb{C}^3$, so $\det DP$ is a polynomial with no complex zero.

A nonconstant polynomial over $\mathbb{C}$ has a zero after restriction to a suitable complex line. Hence every nowhere-zero polynomial on $\mathbb{C}^3$ is constant. At the origin, the displayed linear terms give

$$DP(0,0,0) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & -2 \end{pmatrix}.$$

Consequently,

$$\det DP \equiv -1.$$

The large cancellation in the expanded Jacobian is therefore not accidental: it is forced by local fidelity, polynomial closure, and the fact that a nowhere-zero complex polynomial is a unit.

6 Compact certificate

Introduce source coordinates (x, y, t) by

$$(a, y, z) = \left(x, y, -\frac{1}{2}t\right)$$

and apply the linear target change

$$S(u, v, w) = (w, 2v, 2u).$$

Define

$$F = S \circ P \circ T, \qquad T(x,y,t) = \left(x, y, -\frac{1}{2}t\right).$$

Expanding gives the compact certificate

$$F_1(x,y,t) = (1+xy)^3 t + y^2(1+xy)(4+3xy),$$

$$F_2(x,y,t) = y + 3x(1+xy)^2 t + 3xy^2(4+3xy),$$

$$F_3(x,y,t) = 2x - 3x^2 y - x^3 t.$$

Since

$$\det DS = -4, \qquad \det DP = -1, \qquad \det DT = -\frac{1}{2},$$

the chain rule gives

$$\det DF = (-4)(-1)\left(-\frac{1}{2}\right) = -2.$$

It remains to expose the global alias. Let

$$p_0 = \left(0, 0, -\frac{1}{4}\right), \qquad p_\varepsilon = \left(\varepsilon, -\frac{3}{2}\varepsilon, \frac{13}{2}\right), \quad \varepsilon \in \{1, -1\}.$$

At p_0, direct substitution gives $F(p_0) = (-\frac{1}{4}, 0, 0)$. At p_ε,

$$xy = -\frac{3}{2}, \qquad 1 + xy = -\frac{1}{2}, \qquad y^2 = \frac{9}{4}.$$

Therefore

$$F_1(p_\varepsilon) = -\frac{13}{16} + \frac{9}{16} = -\frac{1}{4},$$

$$F_2(p_\varepsilon) = -\frac{3\varepsilon}{2} + \frac{39\varepsilon}{8} - \frac{27\varepsilon}{8} = 0,$$

$$F_3(p_\varepsilon) = 2\varepsilon + \frac{9\varepsilon}{2} - \frac{13\varepsilon}{2} = 0.$$

Thus the three distinct points satisfy

$$F(p_0) = F(p_1) = F(p_{-1}) = \left(-\frac{1}{4}, 0, 0\right).$$

The map F is polynomial and has constant nonzero Jacobian determinant, yet it is not injective. This proves Theorem 1. For $n > 3$, the product map

$$(F, \mathrm{id}_{\mathbb{C}^{n-3}}) : \mathbb{C}^n \longrightarrow \mathbb{C}^n$$

has the same collision and the same nonzero constant Jacobian up to the identity factor.

7 Closure audit

The UNIVERSAL PROOF SELECTION NETWORK requires closure before presentation. The proof satisfies the audit.

- **Exact claim:** an explicit counterexample over $\mathbb{C}^3$ was constructed.
- **Domain:** every displayed map is polynomial on the stated affine space.
- **Local condition:** $\det DF \equiv -2$, established structurally by the chain rule.
- **Global failure:** three distinct rational points have one exact image.
- **Gauge control:** the continuous scaling ambiguity was removed before the discrete collision was used.
- **Boundary case:** $a = 0$ was handled separately; no illegal division by a occurs.
- **Circularity:** affineness, local fidelity, and noninjectivity were proved by separate obligations.
- **Theorem/experiment separation:** symbolic computation was used only as an independent exact audit; the written proof does not depend on numerical evidence.

8 Interpretation

The counterexample has a compact conceptual form.

> **Local differential fidelity does not imply global state fidelity.**

The Jacobian records whether the visible representation collapses infinitesimal directions. It cannot record a discrete state label that the representation has already erased. Here that label is the choice of which root belongs to the marked linear factor.

The resultant normalization proves that no continuous ambiguity remains. The affine slice proves that the constrained state space is still ordinary three-dimensional affine space. The collision proves that the erased discrete label remains globally consequential. The final polynomial is the compressed certificate of that mechanism.

Provenance and disclosure

The explicit degree-seven counterexample displayed above was publicly attributed to the Fable AI system in July 2026 and was subsequently discussed in geometric form by Terence Tao and David Speyer [1, 2]. This note does **not** claim discovery of the displayed map. It gives an independently organized proof through Jon Poplett's established pipeline.

representation $\longrightarrow$ constraint $\longrightarrow$ local control $\longrightarrow$ collision
$\longrightarrow$ independent witness $\longrightarrow$ boundary closure $\longrightarrow$ exact certificate

- **Primary author and investigator:** Jon Poplett.
- **Assistance:** AI-assisted drafting, algebra checking, and document preparation.
- **Verification:** all decisive polynomial identities were checked exactly over the rational numbers; no floating-point inference enters the theorem.

References

[1] Terence Tao, *A digestion of the Jacobian conjecture counterexample*, What's New, 21 July 2026. Online article.

[2] David E. Speyer, *The new counterexample to the Jacobian conjecture*, Secret Blogging Seminar, 20 July 2026. Online article.

[3] Jon Poplett, *Universal Proof Selection Network—Proof Construction Protocol*, unpublished working protocol, July 2026.

[4] Jon Poplett, *Checkpoint Process Reports 01–11: Finite Local Neighborhood Aliasing*, unpublished research notes, 13 June 2026.